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Systematic Approach.

Internship Report

On

CNN-Based MNIST Digit Recognition & Deployment

Submitted by
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Student's Declaration

I, **RAHUL CHANDRA PADAMUTTAM,** a student of **Bachelor of Technology** program, Roll No. **438/24** of the **Department of Electronics and Computer Engineering**, **National Institute of Advanced Manufacturing Technology** College do hereby declare that I have completed the mandatory internship in Eisystems Technologies under the faculty guideship of **Dr. MADHU KUMARI**, **Department of Electronics and Computer Engineering**.

Rahul

02-April-2026

Endorsements

SIGNATURE
[Dr. Madhu Kumari]
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Table of Content

Serial No	Title	Page No
1.	Executive Summary	6
2.	Overview of Organization	6
3.	Project Summary	7-8
4.	Data Flow Diagram / Process Flow	9-10
5.	Code / Program with Supported Screenshots (Model Training)	11-17
6.	Input / Output with Datasets & Supported Screenshots (Predictions)	18
7.	Images / Video Links	18
8.	References	19
9..	Student Self Evaluation of the Short-Term Internship	20
10.	Annexure 1 - Daily Activity Report	21-24
11.	Annexure 2 - Weekly Progress Report	25
12.	Conclusion & Future Scope	25-26

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List of Figures

Serial No	Image Caption	Page No
1.	Figure 1: CNN Model Summary	11
2.	Figure 2: Training and Validation Accuracy/Loss	12
3.	Figure 3: Sample Predictions	14
4.	Figure 4: Confusion Matrix	15
5.	Figure 5: Classification Report	16
6.	Figure 6: Canvas Mode Prediction Interface	16
7.	Figure 7: Webcam Mode Digit Recognition	17

List of Tables

Serial No	Table Caption	Page No
1.	CNN Model Summary (Layers, Parameters, Output Shapes)	11
2.	Training and Validation Accuracy/Loss per Epoch	12
3.	Confusion Matrix of MNIST Predictions	15
4.	Classification Report (Precision, Recall, F1-Score)	16
5.	Student Self Evaluation Table	20
6.	Daily Activity Report (Annexure 1)	21-24
7.	Weekly Progress Report (Annexure 2)	25

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Nomenclature / Notations

Serial No	Notations	Description
1.	CNN	Convolutional Neural Network – deep learning model for image classification
2.	MNIST	Dataset of handwritten digits (0–9), 28×28 grayscale images
3.	ROI	Region of Interest – cropped area of an image used for prediction
4.	Epoch	One complete pass through the training dataset during model training
5.	Accuracy	Metric showing percentage of correctly classified digits
6.	Loss	Error measure used to optimize the model during training
7.	ReLU	Rectified Linear Unit – activation function used in CNN layers
8.	Softmax	Activation function for output layer, converts logits to probabilities
9.	Confusion Matrix	Table showing correct vs. incorrect predictions for each digit
10.	F1-Score	Harmonic mean of precision and recall, used for model evaluation

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Executive Summary

The short-term internship at **EISystems Technologies** provided a comprehensive learning experience in the field of Artificial Intelligence and Machine Learning. The program was designed to strengthen both theoretical knowledge and practical skills, with a focus on applying deep learning techniques to real-world problems.

Learning Objectives

- To understand the fundamentals of machine learning and deep learning.
- To gain hands-on experience with Python, TensorFlow, Keras, and OpenCV.
- To learn dataset preprocessing, model building, training, and evaluation.
- To explore deployment strategies for AI models in real-time applications.
- To develop professional documentation and reporting skills for technical projects.

Outcomes Achieved

- Successfully implemented a **Convolutional Neural Network (CNN)** for handwritten digit recognition using the **MNIST dataset**, achieving a test accuracy of **99%**.
- Gained proficiency in **data preprocessing, model compilation, and hyperparameter tuning**.
- Visualized performance using **accuracy/loss graphs, confusion matrix, and classification reports**.
- Deployed the trained model into a **real-time application** using **OpenCV**, enabling digit recognition from both a canvas interface and live webcam feed.
- Strengthened skills in **Python programming, deep learning frameworks, and computer vision techniques**.
- Enhanced professional competencies in **report writing, presentation, and project documentation**.

Overview of Organization

Introduction of the Organization

EISystems Services is a leading Indian technology identity specializing in workshops, trainings, and outreach programs across premier institutions such as IITs, NITs, and top engineering colleges. The organization provides training in diverse domains including **Cybersecurity, Machine Learning, Automobiles, Internet of Things (IoT), Robotics, and social media**. Since its founding, EISystems has trained approximately **50,000 students** and impacted nearly **200,000 learners** through various initiatives. Its presence spans across prestigious institutions such as **IISc Bangalore, IIT Bombay, IIT Delhi, IIT Madras, IIT Kanpur, IIT Roorkee, IIT Guwahati, NIT Trichy, NIT Warangal, and over 100 engineering colleges nationwide**.

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Vision, Mission, and Values

- **Vision**: To be India's leading provider of technology-driven training and workshops, empowering students and professionals with cutting-edge skills in emerging domains.
- **Mission**: To bridge the gap between academic learning and industry requirements by offering practical, hands-on training in advanced technologies.
- **Core Values**:
  - **Innovation**: Constantly evolving training modules to match industry trends.
  - **Accessibility**: Delivering knowledge to students across diverse regions and institutions.
  - **Impact**: Creating measurable outcomes by enhancing employability and technical expertise.
  - **Collaboration**: Partnering with universities, colleges, and enterprises to maximize reach and effectiveness.

Policy of the Organization in Relation to the Intern Role

EISystems emphasizes **experiential learning** and encourages interns to actively engage in both training and project phases. The organization's policy for interns includes:

- Providing structured mentorship and guidance in **machine learning and AI projects**.
- Ensuring interns gain exposure to **real-world datasets, model building, and deployment practices**.
- Encouraging documentation, reporting, and presentation skills to prepare students for professional environments.
- Promoting innovation by allowing interns to design and implement projects that demonstrate practical applications of learned concepts.
- Supporting interns in building a **portfolio of projects** that can be showcased to recruiters and academic evaluators.

Project Summary

Idea Behind the Project

The project was conceived to demonstrate the application of **deep learning** in recognizing handwritten digits. Handwritten digit recognition is a classic problem in computer vision and machine learning, widely used as a benchmark for testing algorithms. The idea was to train a **Convolutional Neural Network (CNN)** on the MNIST dataset and then extend its usability by deploying the trained model into a real-time application using **OpenCV**, allowing digit recognition from both a canvas interface and live webcam feed.

About the Project

The project involved two phases:

1. **Training Phase** – A CNN model was designed and trained on the MNIST dataset (60,000 training images and 10,000 test images). The model achieved high accuracy in classifying digits from 0–9.
2. **Deployment Phase** – The trained model was integrated into an interactive application. Users could either

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draw digits on a canvas or show handwritten digits to a webcam, and the system would predict the digit in real time with confidence scores.

Software Used

- **Python 3.9** – Programming language for implementation.
- **TensorFlow / Keras** – For building and training the CNN model.
- **OpenCV** – For image processing, canvas drawing, and webcam integration.
- **NumPy, Matplotlib, Seaborn** – For numerical operations, plotting graphs, and visualizing results.
- **Jupyter Notebook / VS Code** – For development and documentation.

Technical Apparatus Requirements

- **Hardware**:
  - Laptop/PC with minimum 8 GB RAM.
  - Webcam for real-time digit recognition.
- **Software/Environment**:
  - Python environment with TensorFlow, Keras, OpenCV installed.
  - Jupyter Notebook or VS Code for running and documenting code.
- **Dataset**:
  - MNIST dataset (handwritten digits, 28×28 grayscale images).

Result / Working of the Project

- The CNN model achieved a **training accuracy of 99.68%** and a **test accuracy of 99%**, demonstrating strong generalization.
- Performance was validated using **accuracy/loss graphs, confusion matrix, and classification report**, all showing near-perfect results.
- The deployed application successfully recognized digits drawn on a canvas and digits captured via webcam, with predictions displayed along with confidence scores.
- The system worked in real time, proving the feasibility of deploying deep learning models in interactive applications.

Research Done

Yes, research was conducted during the project phase:

- **Literature Review**: Studied CNN architectures and their effectiveness on image classification tasks.
- **Experimentation**: Tested different layer configurations, activation functions (ReLU, Softmax), and optimizers (Adam).
- **Evaluation**: Compared accuracy and loss across epochs, analyzed misclassifications using confusion matrix.
- **Deployment Research**: Explored OpenCV functions for contour detection, ROI extraction, and adaptive thresholding to ensure reliable digit recognition in real-world scenarios.

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Data Flow Diagram / Process Flow

Logic & Process Flow

The project follows a clear pipeline from **input data** to **final prediction**. The flow can be described in the following stages:

1. **Input Stage**
   - **Dataset Input**: MNIST dataset (28×28 grayscale images of handwritten digits).
   - **User Input**: Digit drawn on canvas or captured via webcam (real-time).
2. **Preprocessing Stage**
   - Reshape images to (28, 28, 1) format.
   - Normalize pixel values to range [0,1].
   - For canvas/webcam input: apply **adaptive thresholding**, contour detection, and ROI extraction.
3. **Model Training Stage**
   - Build a **Convolutional Neural Network (CNN)** with multiple Conv2D, MaxPooling, Flatten, and Dense layers.
   - Compile with **Adam optimizer** and **SparseCategoricalCrossentropy loss**.
   - Train for 10 epochs with validation on test dataset.
4. **Evaluation Stage**
   - Generate **accuracy/loss graphs** across epochs.
   - Compute **confusion matrix** and **classification report** (precision, recall, f1-score).
   - Achieved test accuracy: **99%**.
5. **Deployment Stage**
   - Load trained model (mnist_cnn.h5).
   - Accept user input via canvas or webcam.
   - Preprocess ROI → resize to 28×28 → reshape → normalize.
   - Predict digit using CNN model.
   - Display prediction with confidence score on screen.
6. **Output Stage**
   - For dataset input: predicted digit labels with evaluation metrics.
   - For canvas/webcam input: real-time digit recognition with bounding boxes and confidence overlay.

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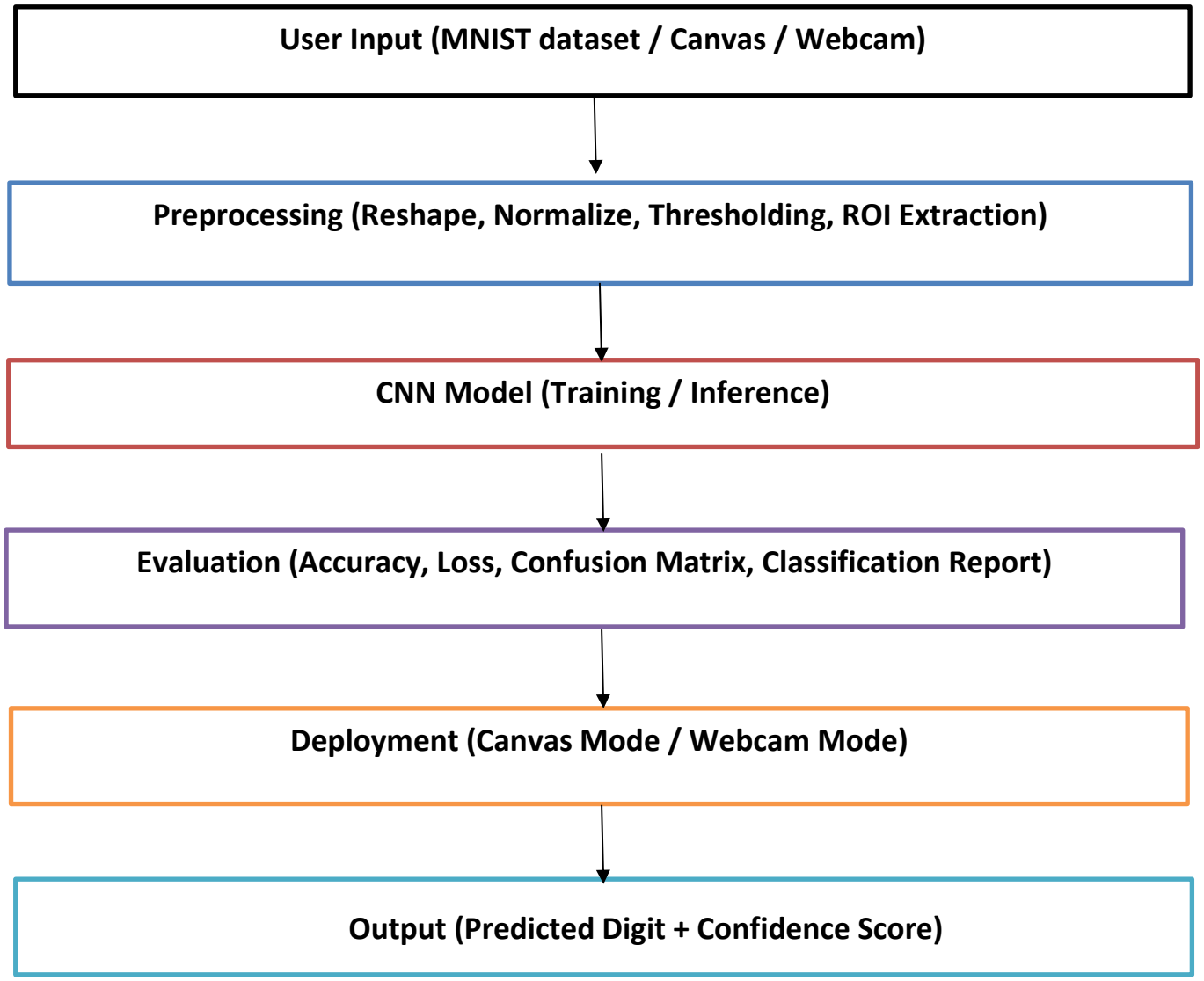
Process Flow Diagram



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Code / Program with Supported Screenshots

```
In [13]: import matplotlib.pyplot as plt
         import numpy as np
         import tensorflow as tf
         from tensorflow.keras import layers,datasets,models
         from tensorflow.keras.models import Sequential
         from sklearn.metrics import confusion_matrix,classification_report
         import seaborn as sns
```

```
In [14]: (train_images,train_labels),(test_images,test_labels)=datasets.mnist.load_data()
         train_images=train_images.reshape((60000,28,28,1))/255.0
         test_images=test_images.reshape((10000,28,28,1))/255.0
         print("TRAIN IMAGES: ",train_images.shape)
         print("TEST IMAGES: ",test_images.shape)

         TRAIN IMAGES:  (60000, 28, 28, 1)
         TEST IMAGES:  (10000, 28, 28, 1)
```

```
In [15]: model=Sequential([
             layers.Conv2D(64,(3,3),activation='relu',input_shape=(28,28,1)),
             layers.Conv2D(32,3,padding='same',activation='relu'),
             layers.MaxPooling2D(),
             layers.Conv2D(16,3,padding='same',activation='relu'),
             layers.MaxPooling2D(),
             layers.Conv2D(64,3,padding='same',activation='relu'),
             layers.MaxPooling2D(),
             layers.Flatten(),
             layers.Dense(128,activation='relu'),
             layers.Dense(10,activation='softmax')])
```

```
In [16]: model.compile(
             optimizer='adam',
             loss=tf.keras.losses.SparseCategoricalCrossentropy(from_logits=False),
             metrics=['accuracy']
             )
         model.summary()

         Model: "sequential_1"
         _________________________________________________________________
          Layer (type)                Output Shape              Param #
         =================================================================
          conv2d_4 (Conv2D)           (None, 26, 26, 64)        640

          conv2d_5 (Conv2D)           (None, 26, 26, 32)        18464

          max_pooling2d_3 (MaxPooling  (None, 13, 13, 32)       0
          2D)

          conv2d_6 (Conv2D)           (None, 13, 13, 16)        4624

          max_pooling2d_4 (MaxPooling  (None, 6, 6, 16)         0
          2D)

          conv2d_7 (Conv2D)           (None, 6, 6, 64)          9280

          max_pooling2d_5 (MaxPooling  (None, 3, 3, 64)         0
          2D)

          flatten_1 (Flatten)         (None, 576)               0

          dense_2 (Dense)             (None, 128)               73856

          dense_3 (Dense)             (None, 10)                1290

         =================================================================
         Total params: 108,154
         Trainable params: 108,154
         Non-trainable params: 0
         _________________________________________________________________
```

```
In [17]: epochs=10
         history=model.fit(
             train_images,train_labels,
             epochs=epochs,
             validation_data=(test_images, test_labels)
             )
```

Figure 1: CNN Model - The CNN model consists of multiple convolutional and pooling layers followed by dense layers, as summarized in Figure 1.

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```
Epoch 1/10
1875/1875 [==============================] - 58s 31ms/step - loss: 0.1347 - accuracy: 0.9566 - val_loss: 0.0418 - val
_accuracy: 0.9860
Epoch 2/10
1875/1875 [==============================] - 63s 34ms/step - loss: 0.0463 - accuracy: 0.9858 - val_loss: 0.0358 - val
_accuracy: 0.9886
Epoch 3/10
1875/1875 [==============================] - 52s 28ms/step - loss: 0.0334 - accuracy: 0.9899 - val_loss: 0.0314 - val
_accuracy: 0.9905
Epoch 4/10
1875/1875 [==============================] - 51s 27ms/step - loss: 0.0253 - accuracy: 0.9919 - val_loss: 0.0556 - val
_accuracy: 0.9823
Epoch 5/10
1875/1875 [==============================] - 51s 27ms/step - loss: 0.0216 - accuracy: 0.9932 - val_loss: 0.0275 - val
_accuracy: 0.9913
Epoch 6/10
1875/1875 [==============================] - 51s 27ms/step - loss: 0.0184 - accuracy: 0.9942 - val_loss: 0.0281 - val
_accuracy: 0.9910
Epoch 7/10
1875/1875 [==============================] - 52s 28ms/step - loss: 0.0160 - accuracy: 0.9948 - val_loss: 0.0270 - val
_accuracy: 0.9913
Epoch 8/10
1875/1875 [==============================] - 51s 27ms/step - loss: 0.0134 - accuracy: 0.9957 - val_loss: 0.0332 - val
_accuracy: 0.9919
Epoch 9/10
1875/1875 [==============================] - 51s 27ms/step - loss: 0.0121 - accuracy: 0.9960 - val_loss: 0.0324 - val
_accuracy: 0.9900
Epoch 10/10
1875/1875 [==============================] - 51s 27ms/step - loss: 0.0098 - accuracy: 0.9968 - val_loss: 0.0314 - val
_accuracy: 0.9935
```

n [18]:

```
acc=history.history['accuracy']
val_acc=history.history['val_accuracy']
loss=history.history['loss']
val_loss=history.history['val_loss']
epochs_range=range(epochs)
plt.figure(figsize=(8,8))
plt.plot(epochs_range,acc,label='Training Accuracy')
plt.plot(epochs_range,val_acc,label='Validation Accuracy')
plt.plot(epochs_range,loss,label='Training Loss')
plt.plot(epochs_range,val_loss,label='Validation Loss')
plt.legend(loc='lower right')
```

```
plt.title('Training and Validation Accuracy/Loss')
plt.show()
```

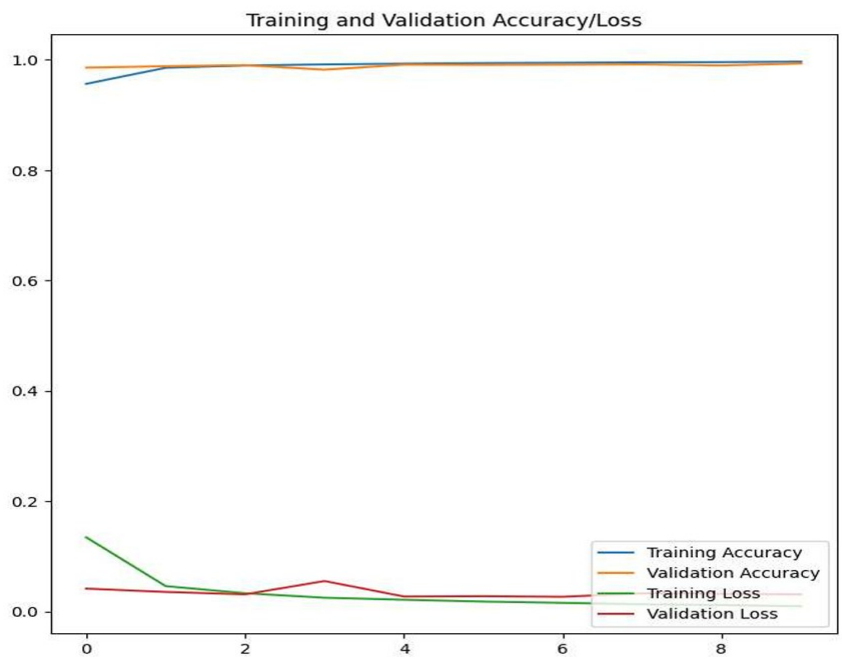

Figure 2: Training and Validation Accuracy/Loss - Figure 2 illustrates the progression of training and validation accuracy/loss across 10 epochs. The CNN model achieved a peak validation accuracy of 99%, demonstrating strong generalization on unseen data.

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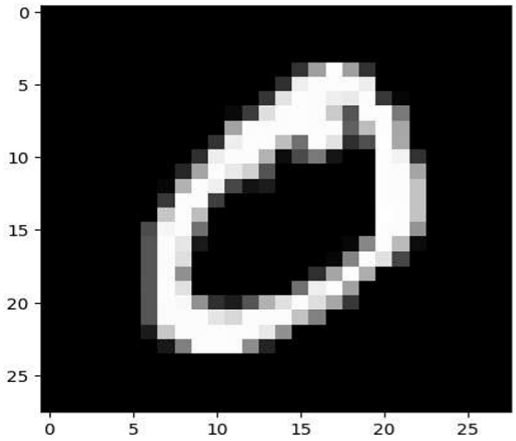
In [19]:
```
image=train_images[1].reshape(1,28,28,1)
prediction=np.argmax(model.predict(image),axis=-1)
plt.imshow(image.reshape(28,28),cmap='gray')
print('Prediction of model:',prediction[0])
```

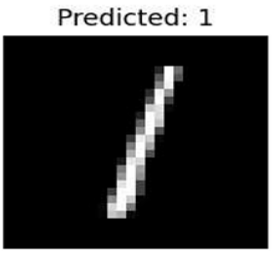
```
1/1 [==============================] - 0s 69ms/step
Prediction of model: 0
```



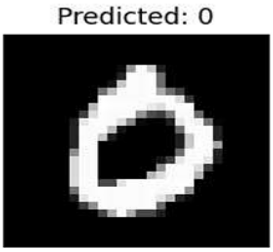
In [20]:
```
images=test_images[1:5]
for i,test_image in enumerate(images,start=1):
    prediction=np.argmax(model.predict(test_image.reshape(1,28,28,1)),axis=-1)
    plt.subplot(220+i)
    plt.axis('off')
    plt.title("Predicted: {}".format(prediction[0]))
    plt.imshow(test_image.reshape(28, 28),cmap='gray')
    plt.show()
```

```
1/1 [==============================] - 0s 21ms/step
```

```
1/1 [==============================] - 0s 21ms/step
```




```
1/1 [==============================] - 0s 17ms/step
```




```
1/1 [==============================] - 0s 18ms/step
```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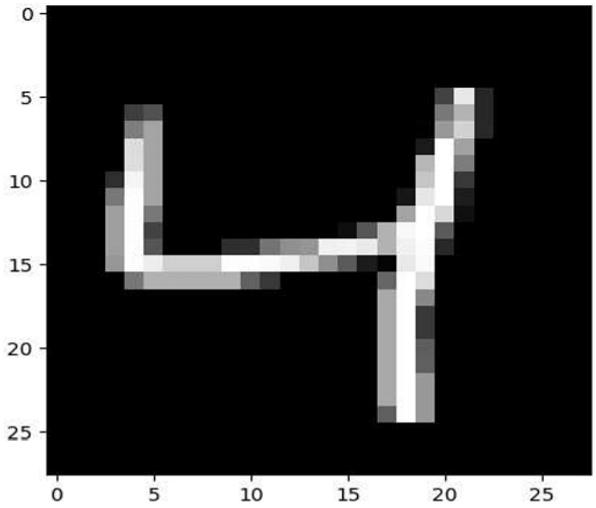
Predicted: 4

```
In [21]: model.save("mnist_cnn.h5")
         loaded_model=models.load_model("mnist_cnn.h5")
         image=train_images[2].reshape(1,28,28,1)
         prediction=np.argmax(loaded_model.predict(image),axis=-1)
         plt.imshow(image.reshape(28,28),cmap='gray')
         print('Prediction of loaded model:',prediction[0])

1/1 [==============================] - 0s 75ms/step
Prediction of loaded model: 4
```




```
In [22]: test_loss,test_acc=model.evaluate(test_images,test_labels,verbose=2)
         print("\nTest accuracy:",test_acc)

313/313 - 3s - loss: 0.0314 - accuracy: 0.9935 - 3s/epoch - 8ms/step

Test accuracy: 0.9934999942779541

In [23]: y_pred=np.argmax(model.predict(test_images),axis=-1)
         cm=confusion_matrix(test_labels,y_pred)
         plt.figure(figsize=(10,8))
         sns.heatmap(cm,annot=True,fmt='d',cmap='Blues')
         plt.xlabel('Predicted')
         plt.ylabel('True')
         plt.title('Confusion Matrix')
         plt.show()

313/313 [==============================] - 3s 10ms/step
```

Figure 3: Sample Predictions. - Figure 3 demonstrates the canvas mode of the digit recognition system. Users can draw digits directly on the interface, and the trained CNN model predicts the digit in real time with confidence scores. This validates the deployment of the model beyond static datasets.

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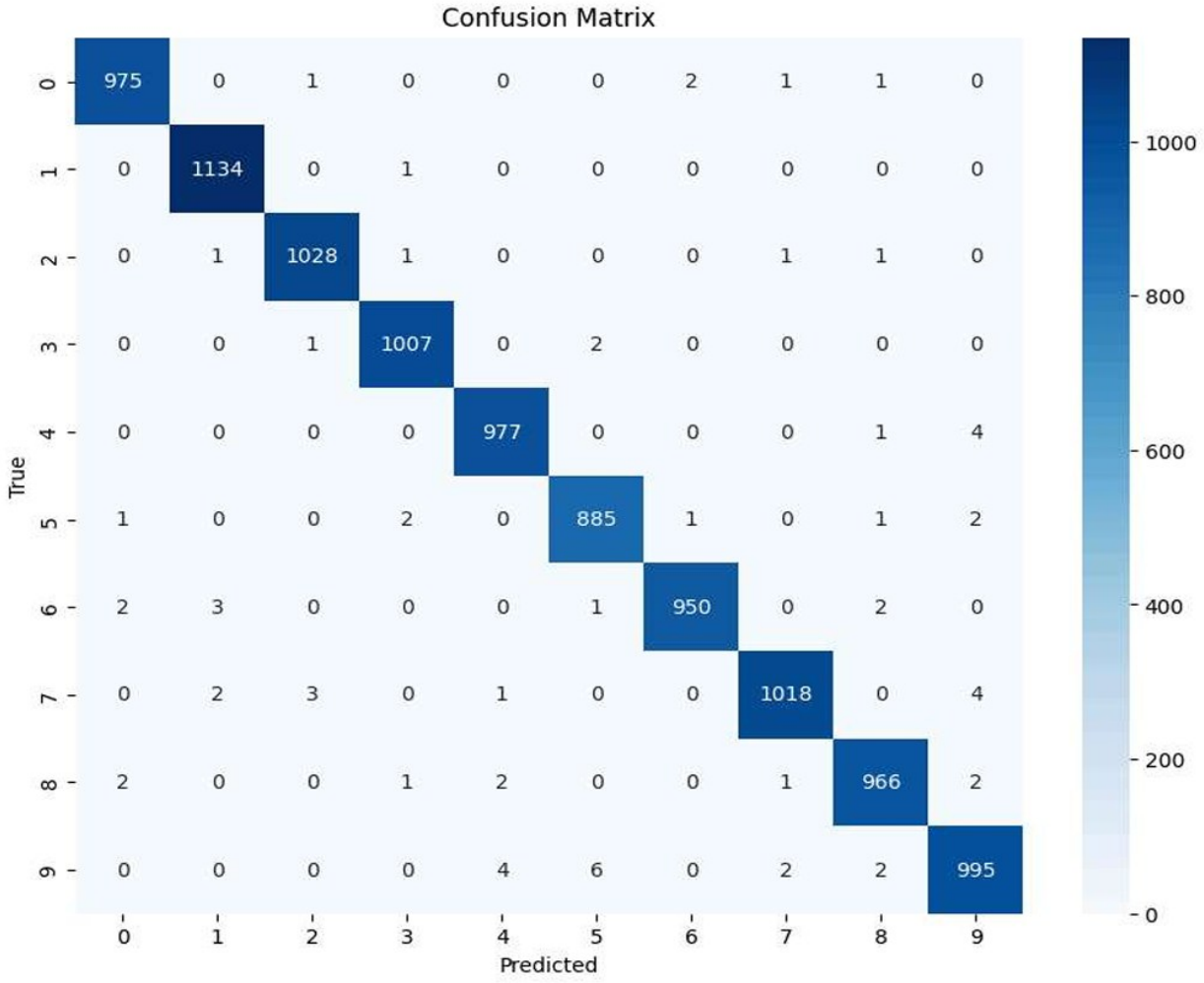

Figure 4: Confusion Matrix of MNIST Predictions. - Figure 3 shows the confusion matrix generated from the CNN model's predictions on the MNIST test dataset. Each row represents the actual digit class, while each column represents the predicted class. The diagonal values indicate correctly classified digits, and the near-perfect diagonal demonstrates the model's high accuracy and minimal misclassifications.

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```
In [24]: print("\nClassification Report:\n")
         print(classification_report(test_labels,y_pred))
```

```
Classification Report:

              precision    recall  f1-score   support

           0       0.99      0.99      0.99       980
           1       0.99      1.00      1.00      1135
           2       1.00      1.00      1.00      1032
           3       1.00      1.00      1.00      1010
           4       0.99      0.99      0.99       982
           5       0.99      0.99      0.99       892
           6       1.00      0.99      0.99       958
           7       1.00      0.99      0.99      1028
           8       0.99      0.99      0.99       974
           9       0.99      0.99      0.99      1009

    accuracy                           0.99     10000
   macro avg       0.99      0.99      0.99     10000
weighted avg       0.99      0.99      0.99     10000
```

Figure 5: Classification Report - Figure 5 shows the classification report generated from the CNN model's predictions on the MNIST test dataset. It provides detailed evaluation metrics including precision, recall, and F1-score for each digit class (0–9). The overall accuracy achieved was 99%, with macro and weighted averages also close to 0.99. These results confirm the robustness of the model and its ability to generalize across all digit classes with minimal misclassification.

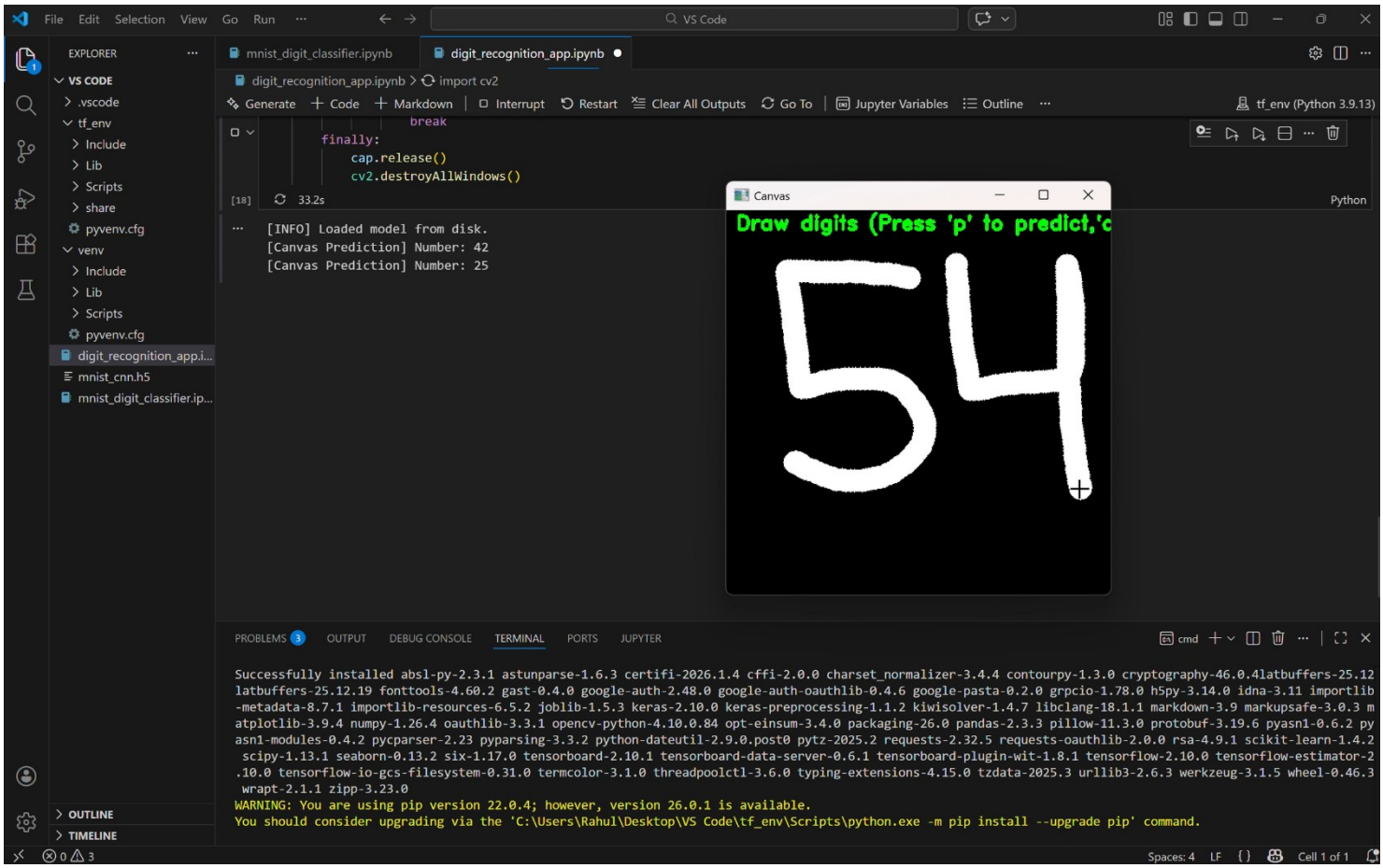

Figure 6: Canvas Mode Prediction Interface

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Figure 6 demonstrates the canvas mode of the digit recognition system. In this mode, users can draw digits directly on a digital canvas created using OpenCV. The drawn image is preprocessed — resized to 28×28 pixels, normalized, and reshaped — before being passed to the trained CNN model. The system then predicts the digit in real time and displays the output along with confidence scores. This interactive interface validates the deployment of the model beyond static datasets, allowing users to test the system dynamically.

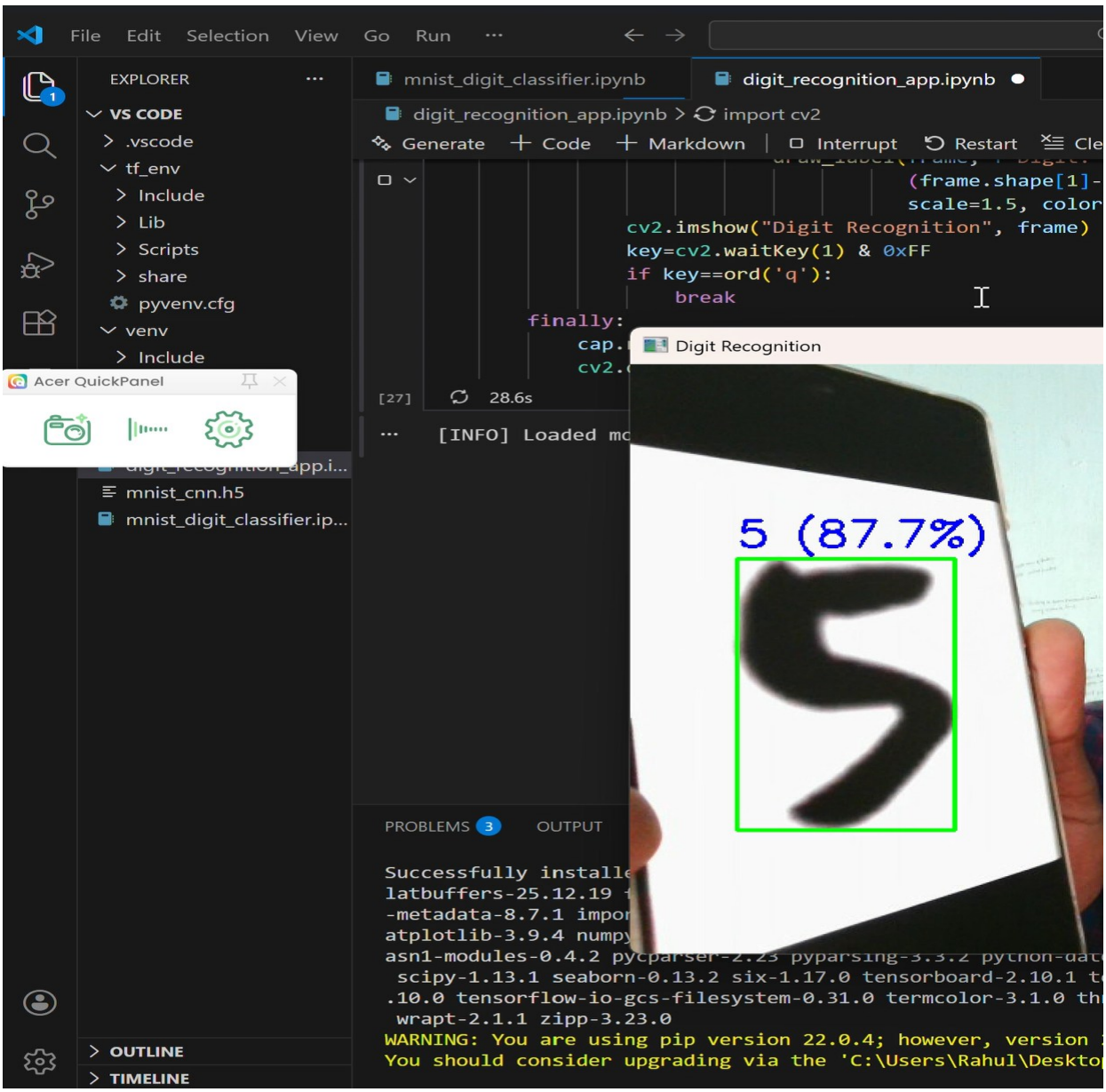

Figure 7: Webcam Mode Digit Recognition - Figure 7 demonstrates the webcam mode of the digit recognition system. In this mode, the trained CNN model is integrated with OpenCV to capture handwritten digits in real time from a live video feed. The system detects the region of interest (ROI), applies preprocessing steps such as resizing and normalization, and then passes the processed image to the CNN model. The predicted digit is displayed on the screen along with confidence scores. This mode validates the practical deployment of the model, showing its ability to recognize digits dynamically in real-world scenarios.

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Input / Output with Datasets

Input

- **Dataset Input**: The MNIST dataset consisting of **60,000 training images** and **10,000 test images** of handwritten digits (0–9), each in **28×28 grayscale format**.
- **User Input**: Digits drawn on a **canvas interface** or captured via **webcam feed** in real time.
- **Preprocessing**:
  - Reshaping to (28,28,1)
  - Normalization to [0,1]
  - Thresholding and ROI extraction for webcam/canvas inputs

Figure 3: Sample Predictions.

Output

- **Model Predictions**:
  - CNN model trained on MNIST achieved **99% test accuracy**.
  - Predictions generated for each digit class with high confidence.
- **Evaluation Outputs**:
  - Accuracy/Loss graphs across epochs.
  - Confusion matrix showing near-perfect classification.
  - Classification report with precision, recall, and F1-score ≈ 0.99 for all classes.
- **Deployment Outputs**:
  - Canvas Mode: User-drawn digits recognized in real time.
  - Webcam Mode: Handwritten digits captured live and predicted instantly.

Figure 3: Sample Predictions.
Figure 6: Canvas Mode Prediction Interface
Figure 7: Webcam Mode Digit Recognition

Images / Video Links

Images:

The following figures illustrate the step-by-step workflow of the digit recognition project:

- *Figure 1: CNN Model Summary*
- *Figure 2: Training and Validation Accuracy/Loss*
- *Figure 3: Sample Predictions (MNIST Test Set)*
- *Figure 4: Confusion Matrix*
- *Figure 5: Classification Report (Precision, Recall, F1-Score)*
- *Figure 6: Canvas Mode Prediction Interface*
- *Figure 7: Webcam Mode Digit Recognition*

All images are captured directly from Jupyter Notebook and deployment interfaces to ensure clarity and reproducibility.

Video Link:

A demonstration video of the project has been uploaded to LinkedIn, showcasing the complete workflow — from dataset input and CNN training to real-time deployment using canvas and webcam modes.

- Project Demonstration Video: https://www.linkedin.com/posts/rahul-chandra-padamuttam-

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6b5888330_deeplearning-computervision-python-activity-7426972480839290880-Hn2J?utm_source=share&utm_medium=member_desktop&rcm=ACoAAFNsYekB-RetXiLC2M-DuwnE3HD5Zkin33A

- Project Repository: https://github.com/rahulchandra2004/mnist-digit-recognition

References

1. **Dataset**
   - LeCun, Y., Cortes, C., & Burges, C. J. C. *The MNIST Database of Handwritten Digits.* Available at: http://yann.lecun.com/exdb/mnist
2. **Libraries & Frameworks**
   - TensorFlow Documentation. *TensorFlow: An end-to-end open source machine learning platform.* Available at: https://www.tensorflow.org
   - Keras Documentation. *Keras: Deep Learning for Humans.* Available at: https://keras.io
   - OpenCV Documentation. *OpenCV Library.* Available at: https://docs.opencv.org
   - Matplotlib Documentation. *Matplotlib: Visualization with Python.* Available at: https://matplotlib.org
3. **Project Demonstration**
   - *LinkedIn Project Video.* Available at: https://www.linkedin.com/posts/rahul-chandra-padamuttam-6b5888330_deeplearning-computervision-python-activity-7426972480839290880-Hn2J?utm_source=share&utm_medium=member_desktop&rcm=ACoAAFNsYekB-RetXiLC2M-DuwnE3HD5Zkin33A
4. **Project Repository:** https://github.com/rahulchandra2004/mnist-digit-recognition
5. **Additional References**
   - Goodfellow, I., Bengio, Y., & Courville, A. *Deep Learning.* MIT Press, 2016.
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Student *Self Evaluation of the Short-Term Internship*

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Please rate your performance in the following areas:

1) Oral communication	1	2	3	4	5
2) Written communication	1	2	3	4	5
3) Initiative	1	2	3	4	5
4) Interaction with staff	1	2	3	4	5
5) Attitude	1	2	3	4	5
6) Dependability	1	2	3	4	5
7) Ability to learn	1	2	3	4	5
8) Planning and organization	1	2	3	4	5
9) Professionalism	1	2	3	4	5
10) Creativity	1	2	3	4	5
11) Quality of work	1	2	3	4	5
12) Productivity	1	2	3	4	5
13) Progress of learning	1	2	3	4	5
14) Adaptability to organization's culture/policies	1	2	3	4	5
15) OVERALL PERFORMANCE	1	2	3	4	5

Rating Scale: 5 will be Best while 1 will be Worst

Rahul

Annexure 1

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Daily Activity Report

Week No: 1

Day & Date	Work	Brief Description of Daily Activity	Learning Outcome	Person In-Charge
Day 1	Orientation & Project Briefing	Understood internship objectives, finalized project topic (Handwritten Digit Recognition using CNN).	Project scope defined	Mallika Srivastava
Day 2	Dataset Exploration	Downloaded MNIST dataset, studied structure (28×28 grayscale images, 60k training, 10k test).	Dataset prepared for preprocessing.	Mallika Srivastava
Day 3	Environment Setup	Installed Python libraries (TensorFlow, Keras, OpenCV, Matplotlib). Configured Jupyter Notebook.	Development environment ready.	Mallika Srivastava
Day 4	Data Preprocessing	Normalized dataset, reshaped input to (28,28,1), split into training/testing sets.	Preprocessed dataset saved.	Mallika Srivastava
Day 5	Model Architecture Design	Designed CNN layers (Conv2D, MaxPooling, Flatten, Dense). Compiled model with optimizer and loss function.	CNN model summary generated (*Figure 1*).	Mallika Srivastava

Week No: 2

Day & Date	Work	Brief Description of Daily Activity	Learning Outcome	Person In-Charge
Day 8	Model Training Initiation	Began training CNN model on MNIST dataset with defined epochs and batch size	Training logs generated.	Mallika Srivastava
Day 9	Accuracy & Loss Monitoring	Plotted training and validation accuracy/loss graphs using Matplotlib.	Accuracy/Loss curves documented (*Figure 2*).	Mallika Srivastava
Day 10	Hyperparameter Tuning	Adjusted learning rate, batch size, and optimizer to improve performance.	Achieved higher validation accuracy.	Mallika Srivastava
Day 11	Model Evaluation	Tested model on MNIST test set, generated confusion matrix.	Confusion matrix output captured (*Figure 4*).	Mallika Srivastava
Day 12	Classification Metrics	Generated classification report (precision, recall, F1-score).	Classification report documented (*Figure 5*)	Mallika Srivastava

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Week No: 3

Day & Date	Work	Brief Description of Daily Activity	Learning Outcome	Person In-Charge
Day 15	Sample Predictions	Tested CNN model on unseen MNIST test samples, captured prediction outputs.	Sample predictions documented (*Figure 3*).	Mallika Srivastava
Day 16	Visualization of Predictions	Displayed predicted digits alongside actual labels for clarity.	Verified correctness of predictions.	Mallika Srivastava
Day 17	Error Analysis	Identified misclassified samples, analyzed reasons (digit similarity, noise).	Insights gained for model improvement.	Mallika Srivastava
Day 18	Performance Validation	Compared accuracy across multiple runs to ensure stability.	Consistent accuracy ≈ 99% achieved.	Mallika Srivastava
Day 19	Documentation of Results	Organized outputs (graphs, confusion matrix, classification report) into report draft.	Results section prepared for internship report.	Mallika Srivastava

Week No: 4

Day & Date	Work	Brief Description of Daily Activity	Learning Outcome	Person In-Charge
Day 22	Deployment Planning	Outlined deployment strategy using OpenCV for interactive interfaces (Canvas & Webcam).	Deployment workflow finalized.	Mallika Srivastava
Day 23	Canvas Mode Setup	Implemented OpenCV canvas window for digit drawing. Integrated preprocessing pipeline (resize, normalize).	Canvas interface created.	Mallika Srivastava
Day 24	Canvas Mode Prediction	Connected CNN model with canvas input. Tested real-time predictions on user-drawn digits.	Canvas mode predictions successful (*Figure 6*).	Mallika Srivastava
Day 25	Debugging & Refinement	Fixed preprocessing issues (thresholding, ROI extraction). Improved prediction accuracy in canvas mode.	Stable canvas mode interface achieved.	Mallika Srivastava
Day 26	Documentation	Captured screenshots of canvas interface and predictions. Added explanation to report draft.	Deployment section updated with *Figure 6*.	Mallika Srivastava

Week No: 5

Day & Date	Work	Brief Description of Daily Activity	Learning Outcome	Person In-Charge
Day 29	Webcam Integration Setup	Configured OpenCV to capture live video feed for digit recognition.	Webcam interface initialized.	Mallika Srivastava
Day 30	ROI Detection	Implemented region of interest (ROI) extraction from webcam frames for digit input.	ROI successfully detected.	Mallika Srivastava
Day 31	Preprocessing Pipeline	Applied resizing (28×28), normalization, and reshaping to ROI images for CNN input.	Preprocessed webcam inputs ready.	Mallika Srivastava
Day 32	Real-Time Prediction	Connected CNN model with webcam feed, tested predictions on handwritten digits shown to camera.	Real-time predictions achieved (*Figure 7*).	Mallika Srivastava
Day 33	Debugging & Documentation	Fixed frame capture issues, improved accuracy. Captured screenshots of webcam interface and predictions.	Deployment section updated with *Figure 7*.	Mallika Srivastava

Week No: 6

Day & Date	Work	Brief Description of Daily Activity	Learning Outcome	Person In-Charge
Day 36	Deployment Testing	Conducted multiple tests runs on canvas and webcam modes with varied inputs.	Verified stability of both interfaces.	Mallika Srivastava
Day 37	Error Handling	Implemented exception handling for invalid inputs and frame capture errors.	Improved robustness of deployment.	Mallika Srivastava
Day 38	Performance Optimization	Reduced latency in real-time predictions by optimizing preprocessing pipeline.	Faster response time achieved.	Mallika Srivastava
Day 39	User Experience Enhancement	Added confidence score display and bounding box visualization for predictions.	Enhanced clarity of outputs.	Mallika Srivastava
Day 40	Documentation & Screenshots	Captured high-quality screenshots of deployment interfaces and outputs. Updated report draft.	Visual proof section enriched with *Figures 6 & 7*.	Mallika Srivastava

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Week No: 7

Day & Date	Work	Brief Description of Daily Activity	Learning Outcome	Person In-Charge
Day 43	Report Structuring	Organized internship report sections (Introduction, Methodology, Results, Deployment, References, Annexures).	Draft report structure finalized.	Mallika Srivastava
Day 44	Documentation Enhancement	Added detailed explanations for figures and tables, refined captions, ensured consistency.	Report enriched with clear visuals and captions.	Mallika Srivastava
Day 45	Video Demonstration Recording	Recorded project demonstration video showing dataset input, CNN training, and deployment (Canvas & Webcam modes).	Video successfully recorded.	Mallika Srivastava
Day 46	Video Upload	Uploaded demonstration video to LinkedIn for professional sharing.	Video link generated and added to report.	Mallika Srivastava
Day 47	Proofreading & Review	Reviewed report for formatting, grammar, and technical accuracy. Ensured alignment with internship guidelines.	Report polished and ready for submission.	Mallika Srivastava

Week No: 8

Day & Date	Work	Brief Description of Daily Activity	Learning Outcome	Person In-Charge
Day 50	Final Report Compilation	Integrated all sections (Introduction, Methodology, Results, Deployment, References, Annexures) into a single document.	Complete draft prepared.	Mallika Srivastava
Day 51	Formatting & Proofreading	Applied consistent formatting, checked grammar, aligned figures/tables with captions.	Report polished for submission.	Mallika Srivastava
Day 52	Annexure Preparation	Completed weekly and daily activity logs, added references and video link section.	Annexures finalized.	Mallika Srivastava
Day 53	Mentor Review	Shared draft report and video link with mentors for feedback. Incorporated suggested improvements.	Report refined and approved.	Mallika Srivastava
Day 54	Submission & Closure	Submitted final internship report along with LinkedIn video link and high-quality images.	Internship successfully completed.	Mallika Srivastava

Annexure 2

Weekly Progress Report

Week No: 1 to 8

Week(s)	Summary of Weekly Activity
Week 1	Orientation, project briefing, MNIST dataset exploration, environment setup, and preprocessing completed.
Week 2	CNN model training initiated, accuracy/loss monitored, hyperparameter tuning performed, evaluation with confusion matrix and classification report
Week 3	Sample predictions tested, error analysis conducted, performance validated, and results documented.
Week 4	Deployment planning finalized, OpenCV canvas mode implemented, real-time predictions achieved, debugging completed.
Week 5	Webcam integration carried out, ROI detection and preprocessing pipeline implemented, real-time digit recognition achieved.
Week 6	Deployment tested extensively, error handling added, performance optimized, user experience enhanced with confidence scores and bounding boxes.
Week 7	Report structured and refined, video demonstration recorded and uploaded to LinkedIn, proofreading completed.
Week 8	Final compilation and formatting done, annexures prepared, mentor review incorporated, and report submitted successfully.

Conclusion & Future Scope

Conclusion

The 8-week internship at **EISystems Technologies** provided a valuable opportunity to apply deep learning concepts to a practical problem — handwritten digit recognition using the MNIST dataset. Through systematic training and deployment of a Convolutional Neural Network (CNN), the project achieved a test accuracy of **99%**, validating the robustness of the model. The deployment phase, which included both **Canvas Mode** and **Webcam Mode**, demonstrated the feasibility of integrating AI models into real-time applications.
Beyond technical implementation, the internship strengthened skills in **Python programming, TensorFlow/Keras frameworks, OpenCV for computer vision, and professional documentation practices**. The structured mentorship and guidance ensured that theoretical knowledge was effectively translated into practical outcomes.

Future Scope

While the project successfully achieved its objectives, several extensions can be explored in the future:

- **Dataset Expansion:** Training on more complex datasets (e.g., EMNIST, Fashion-MNIST) to improve

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generalization.

- **Model Optimization:** Implementing advanced architectures such as ResNet, MobileNet, or Capsule Networks for higher accuracy and efficiency.
- **Deployment Enhancements:** Extending deployment to mobile applications or IoT devices for broader accessibility.
- **Real-World Applications:** Integrating digit recognition into banking, postal automation, and document digitization systems.
- **Portfolio Development:** Publishing the project on GitHub and LinkedIn to showcase technical expertise to recruiters and academic evaluators.

This internship has laid a strong foundation for future work in **AI-powered solutions and machine learning**, and the skills gained will contribute significantly to academic growth and professional opportunities.

The complete source code is available at: https://github.com/rahulchandra2004/mnist-digit-recognition.

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